

Volumes by the Disc and Washer Methods



The disc method (about the x-axis)

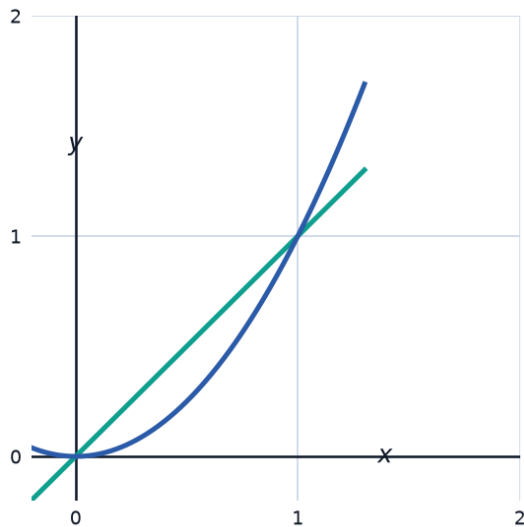
- 1 Find the volume of the solid formed by revolving $y = \sqrt{x}$ about the x-axis on $[0, 4]$.
 - 2 Find the volume of the solid formed by revolving $y = x^2$ about the x-axis on $[0, 2]$.
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The disc method (about the y-axis)

- 3 Find the volume formed by revolving $x = y^2$ about the y-axis on $y \in [0, 2]$.
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The washer method

- 4 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x-axis.



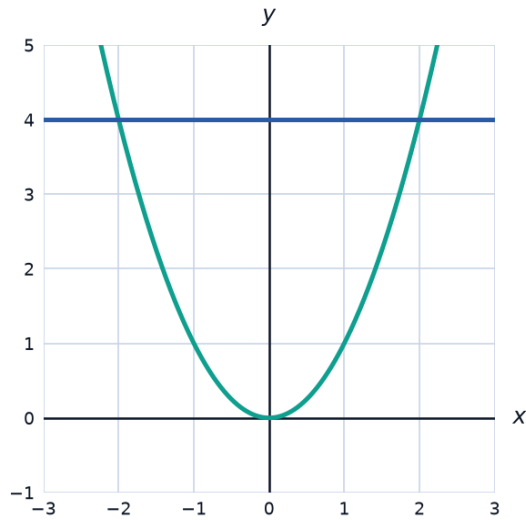
- 5 Find the volume formed by revolving the region between $y = \sqrt{x}$ and $y = x$ about the x-axis on $[0, 1]$.
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Volumes about a horizontal or vertical line (not an axis)

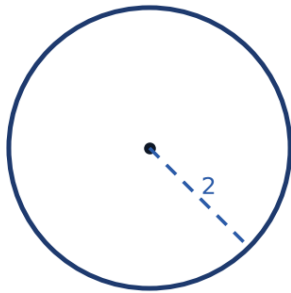
- 6 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.
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Volumes with known cross-sections

- 7 The base of a solid is the region between $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.



- 8 The base of a solid is a circle of radius 2 ($x^2 + y^2 = 4$). Cross-sections perpendicular to the x -axis are semicircles. Find the volume.



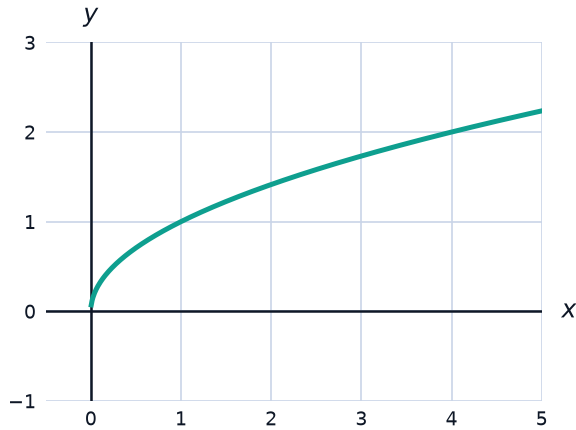
- 9 Set up (but do not evaluate) the integral for the volume of the solid whose base is the region bounded by $y = \sin x$ and the x -axis on $[0, \pi]$, with cross-sections perpendicular to the x -axis that are equilateral triangles.

Volume by revolving about the y -axis (washer method)

- 10 Find the volume formed by revolving the region between $x = y^2$ and $x = 2y$ about the y -axis.
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Visualizing a solid of revolution's generating region

- 11 The graph shows $y = \sqrt{x}$ on $[0, 4]$ from the first disc-method problem. Sketching this region mentally revolved about the x -axis produces a solid — use the volume 8π found earlier to find its average cross-sectional area over the interval, and compare that to the cross-sectional area at the midpoint $x = 2$.



A capstone washer-method problem

- 12 Find the volume formed by revolving the region between $y = 4 - x^2$ and $y = x^2$ about the x -axis (using the region where $4 - x^2 \geq x^2$).

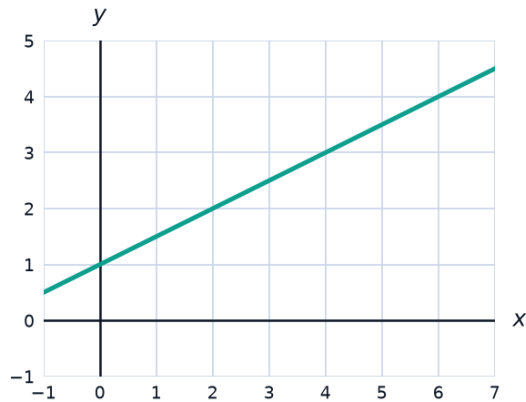
Volumes with rectangular cross-sections

- 13 The base of a solid is the region bounded by $y = x^2$ and $y = 1$. Cross-sections perpendicular to the x -axis are rectangles with height twice the base. Find the volume.

A capstone volume application

- 14 A bowl is formed by revolving $y = \frac{x^2}{4}$ (for $0 \leq x \leq 4$, where x is horizontal distance and y is depth from the rim) about the y -axis. Set up (do not evaluate) the disc-method integral for the volume of the bowl in terms of y , using the fact that $x^2 = 4y$.
- 15 Find the volume formed by revolving $y = e^x$ about the x -axis on $[0, 1]$.

- 16 A vase is designed by revolving $y = \frac{1}{2}x + 1$ about the x -axis on $[0, 6]$, where x is the height (inches) and y is the radius. Find the volume of the vase.



- 17 For the region between $y = x$ and $y = x^3$ on $[0, 1]$ (where $y = x \geq y = x^3$), complete each step for the solid formed by revolving this region about the x -axis:
- a Identify the outer radius R and inner radius r .
 - b Set up the washer-method integral.
 - c Evaluate the integral.

Multiple choice — easy 15

- 18 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 5]$.
- a. 626π
 - b. 125π
 - c. 625π
 - d. $\frac{3125\pi}{6}$
- 19 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 4]$.
- a. 2048π
 - b. $\frac{16391\pi}{7}$
 - c. $\frac{16384\pi}{7}$
 - d. $\frac{4096\pi}{7}$

20 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 6]$.

a. $\frac{46656\pi}{7}$

b. $\frac{279943\pi}{7}$

c. 34992π

d. $\frac{279936\pi}{7}$

21 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 6]$.

a. $\frac{1296\pi}{5}$

b. $\frac{7781\pi}{5}$

c. 1296π

d. $\frac{7776\pi}{5}$

22 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 3]$.

a. $\frac{2187\pi}{8}$

b. $\frac{729\pi}{7}$

c. $\frac{2187\pi}{7}$

d. $\frac{2194\pi}{7}$

23 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 5]$.

a. $\frac{128\pi}{3}$

b. $\frac{125\pi}{3}$

c. $\frac{25\pi}{3}$

d. $\frac{125\pi}{4}$

24 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 3]$.

a. $\frac{248\pi}{5}$

b. $\frac{81\pi}{2}$

c. $\frac{81\pi}{5}$

d. $\frac{243\pi}{5}$

25 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 2]$.

a. $\frac{32\pi}{5}$

b. $\frac{16\pi}{3}$

c. $\frac{16\pi}{5}$

d. $\frac{37\pi}{5}$

26 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 2]$.

a. 2π

b. $\frac{4\pi}{3}$

c. $\frac{8\pi}{3}$

d. $\frac{11\pi}{3}$

27 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 2]$.

a. $\frac{128\pi}{7}$

b. $\frac{135\pi}{7}$

c. $\frac{64\pi}{7}$

d. 16π

28 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 6]$.

a. 54π

b. 72π

c. 73π

d. 12π

29 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 1]$.

a. $\frac{\pi}{3}$

b. $\frac{\pi}{4}$

c. $\frac{5\pi}{6}$

d. $\frac{4\pi}{3}$

30 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 1]$.

a. $\frac{6\pi}{5}$

b. $\frac{\pi}{5}$

c. $\frac{\pi}{6}$

d. $\frac{7\pi}{10}$

31 Find the volume formed by revolving $y = x^4$ about the x -axis on $[0, 2]$.

a. $\frac{512\pi}{9}$

b. $\frac{521\pi}{9}$

c. $\frac{256\pi}{5}$

d. $\frac{256\pi}{9}$

32 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 4]$.

- a. $\frac{16\pi}{3}$
- b. $\frac{67\pi}{3}$
- c. 16π
- d. $\frac{64\pi}{3}$

Multiple choice — medium 15

33 Find the volume formed by revolving the region between $y = x^4$ and $y = x^8$ (on $[0, 1]$) about the x -axis.

- a. $\frac{8\pi}{153}$
- b. $\frac{4\pi}{45}$
- c. $-\frac{8\pi}{153}$
- d. $\frac{26\pi}{153}$

34 Find the volume formed by revolving the region between $y = x^4$ and $y = x^6$ (on $[0, 1]$) about the x -axis.

- a. $-\frac{4\pi}{117}$
- b. $\frac{4\pi}{117}$
- c. $\frac{2\pi}{35}$
- d. $\frac{22\pi}{117}$

35 Find the volume formed by revolving the region between $y = x$ and $y = x^4$ (on $[0, 1]$) about the x -axis.

- a. $\frac{3\pi}{10}$
- b. $\frac{2\pi}{9}$
- c. $-\frac{2\pi}{9}$
- d. $\frac{4\pi}{9}$

- 36 Find the volume formed by revolving the region between $y = x^3$ and $y = x^7$ (on $[0, 1]$) about the x -axis.
- a. $\frac{8\pi}{105}$
 - b. $\frac{22\pi}{105}$
 - c. $\frac{\pi}{8}$
 - d. $-\frac{8\pi}{105}$
- 37 Find the volume formed by revolving the region between $y = x$ and $y = x^6$ (on $[0, 1]$) about the x -axis.
- a. $\frac{10\pi}{39}$
 - b. $\frac{16\pi}{39}$
 - c. $-\frac{10\pi}{39}$
 - d. $\frac{5\pi}{14}$
- 38 Find the volume formed by revolving the region between $y = x^3$ and $y = x^9$ (on $[0, 1]$) about the x -axis.
- a. $\frac{3\pi}{20}$
 - b. $-\frac{12\pi}{133}$
 - c. $\frac{12\pi}{133}$
 - d. $\frac{26\pi}{133}$
- 39 Find the volume formed by revolving the region between $y = x^2$ and $y = x^4$ (on $[0, 1]$) about the x -axis.
- a. $\frac{4\pi}{45}$
 - b. $\frac{14\pi}{45}$
 - c. $\frac{2\pi}{15}$
 - d. $-\frac{4\pi}{45}$

- 40 Find the volume formed by revolving the region between $y = x^2$ and $y = x^6$ (on $[0, 1]$) about the x -axis.
- a. $\frac{8\pi}{65}$
 - b. $-\frac{8\pi}{65}$
 - c. $\frac{4\pi}{21}$
 - d. $\frac{18\pi}{65}$
- 41 Find the volume formed by revolving the region between $y = x^3$ and $y = x^6$ (on $[0, 1]$) about the x -axis.
- a. $\frac{6\pi}{91}$
 - b. $\frac{20\pi}{91}$
 - c. $\frac{3\pi}{28}$
 - d. $-\frac{6\pi}{91}$
- 42 Find the volume formed by revolving the region between $y = x^4$ and $y = x^7$ (on $[0, 1]$) about the x -axis.
- a. $\frac{2\pi}{45}$
 - b. $-\frac{2\pi}{45}$
 - c. $\frac{3\pi}{40}$
 - d. $\frac{8\pi}{45}$
- 43 Find the volume formed by revolving the region between $y = x^3$ and $y = x^8$ (on $[0, 1]$) about the x -axis.
- a. $\frac{10\pi}{119}$
 - b. $\frac{5\pi}{36}$
 - c. $-\frac{10\pi}{119}$
 - d. $\frac{24\pi}{119}$

- 44 Find the volume formed by revolving the region between $y = x^4$ and $y = x^9$ (on $[0, 1]$) about the x -axis.
- a. $\frac{\pi}{10}$
 - b. $-\frac{10\pi}{171}$
 - c. $\frac{28\pi}{171}$
 - d. $\frac{10\pi}{171}$
- 45 Find the volume formed by revolving the region between $y = x^2$ and $y = x^8$ (on $[0, 1]$) about the x -axis.
- a. $\frac{2\pi}{9}$
 - b. $\frac{22\pi}{85}$
 - c. $-\frac{12\pi}{85}$
 - d. $\frac{12\pi}{85}$
- 46 Find the volume formed by revolving the region between $y = x^3$ and $y = x^4$ (on $[0, 1]$) about the x -axis.
- a. $\frac{\pi}{20}$
 - b. $\frac{2\pi}{63}$
 - c. $\frac{16\pi}{63}$
 - d. $-\frac{2\pi}{63}$
- 47 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x -axis.
- a. $\frac{\pi}{6}$
 - b. $\frac{2\pi}{15}$
 - c. $\frac{8\pi}{15}$
 - d. $-\frac{2\pi}{15}$

Multiple choice — hard 15

- 48 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 16$ about the line $y = 16$.
- a. $\frac{16384\pi}{15}$
 - b. $\frac{8192\pi}{15}$
 - c. $\frac{4096\pi}{15}$
 - d. $\frac{16444\pi}{15}$
- 49 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 9$ about the line $y = 9$.
- a. $\frac{1311\pi}{5}$
 - b. $\frac{648\pi}{5}$
 - c. $\frac{1296\pi}{5}$
 - d. $\frac{432\pi}{5}$
- 50 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.
- a. $\frac{256\pi}{15}$
 - b. $\frac{542\pi}{15}$
 - c. $\frac{1039\pi}{30}$
 - d. $\frac{512\pi}{15}$
- 51 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 25$ about the line $y = 25$.
- a. $\frac{5000\pi}{3}$
 - b. $\frac{10000\pi}{3}$
 - c. $\frac{10015\pi}{3}$
 - d. $\frac{2000\pi}{3}$

52 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 1$ about the line $y = 1$.

a. $\frac{17\pi}{30}$

b. $\frac{31\pi}{15}$

c. $\frac{8\pi}{15}$

d. $\frac{16\pi}{15}$

53 The base of a solid is the region between $y = x^2$ and $y = 36$ on $[-6, 6]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

a. $\frac{6912}{5}$

b. $\frac{41472}{5}$

c. $\frac{20736}{5}$

d. $\frac{41502}{5}$

54 The base of a solid is the region between $y = x^2$ and $y = 25$ on $[-5, 5]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

a. $\frac{5000}{3}$

b. $\frac{10000}{3}$

c. $\frac{2000}{3}$

d. $\frac{10015}{3}$

55 The base of a solid is the region between $y = x^2$ and $y = 16$ on $[-4, 4]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

a. $\frac{8192}{15}$

b. $\frac{4096}{15}$

c. $\frac{16384}{15}$

d. $\frac{16444}{15}$

- 56 The base of a solid is the region between $y = x^2$ and $y = 1$ on $[-1, 1]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.
- a. $\frac{8}{15}$
 - b. $\frac{47}{30}$
 - c. $\frac{16}{15}$
 - d. $\frac{31}{15}$
- 57 The base of a solid is the region between $y = x^2$ and $y = 9$ on $[-3, 3]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.
- a. $\frac{1311}{5}$
 - b. $\frac{648}{5}$
 - c. $\frac{432}{5}$
 - d. $\frac{1296}{5}$
- 58 The base of a solid is the region bounded by $y = x^2$ and $y = 9$ on $[-3, 3]$. Cross-sections perpendicular to the x -axis are rectangles with height 2 times the base. Find the volume.
- a. $\frac{1296}{5}$
 - b. $\frac{2592}{5}$
 - c. $\frac{2607}{5}$
 - d. $\frac{5179}{10}$
- 59 The base of a solid is the region bounded by $y = x^2$ and $y = 1$ on $[-1, 1]$. Cross-sections perpendicular to the x -axis are rectangles with height 2 times the base. Find the volume.
- a. $\frac{49}{30}$
 - b. $\frac{32}{15}$
 - c. $\frac{16}{15}$
 - d. $\frac{47}{15}$

- 60 The base of a solid is the region bounded by $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x -axis are rectangles with height 2 times the base. Find the volume.
- a. $\frac{2033}{30}$
 - b. $\frac{512}{15}$
 - c. $\frac{1054}{15}$
 - d. $\frac{1024}{15}$
- 61 The base of a solid is the region bounded by $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x -axis are rectangles with height 4 times the base. Find the volume.
- a. $\frac{2078}{15}$
 - b. $\frac{1024}{15}$
 - c. $\frac{2048}{15}$
 - d. $\frac{512}{15}$
- 62 The base of a solid is the region bounded by $y = x^2$ and $y = 9$ on $[-3, 3]$. Cross-sections perpendicular to the x -axis are rectangles with height 4 times the base. Find the volume.
- a. $\frac{2592}{5}$
 - b. $\frac{5184}{5}$
 - c. $\frac{1296}{5}$
 - d. $\frac{5199}{5}$